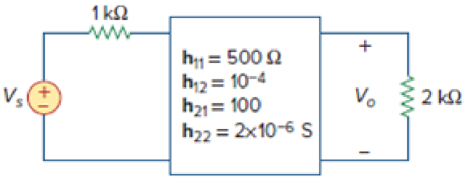
Chapter 19, Problem 56P

Problem

For the network of Fig. 19.107, obtain **Vo / Vs**.

Figure 19.107



Step-by-step solution

Step 1 of 4

Consider the following **h** parameter network:

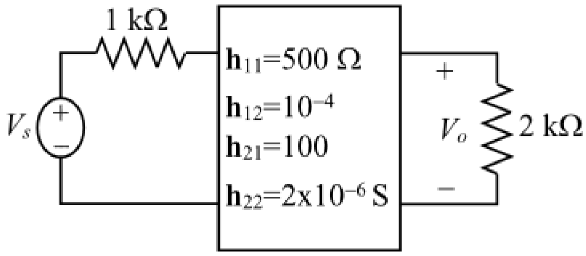


Figure 1

Step 2 of 4

Draw the equivalent **h** parameter circuit.

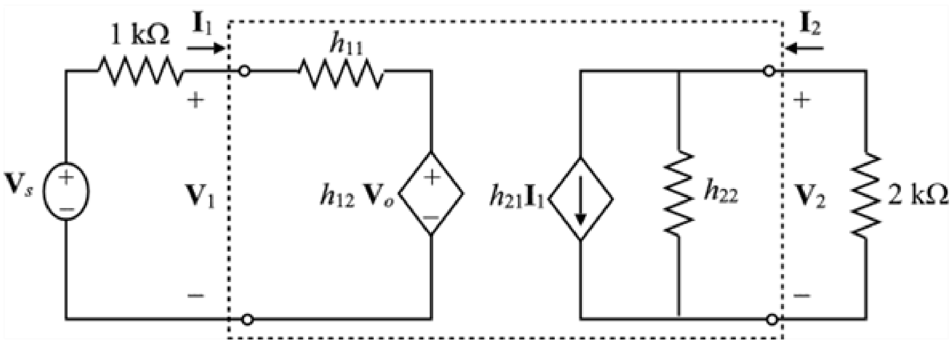


Figure 2: Equivalent h-parameter circuit

Step 3 of 4

Apply Kirchhoff's voltage law to the input loop.

$$-\mathbf{V}_s + 10^3\mathbf{I}_1 + \mathbf{V}_1 = 0$$
$$\mathbf{V}_1 + 1000\mathbf{I}_1 = \mathbf{V}_s$$

Substitute $h_{11}\mathbf{I}_1 + h_{12}\mathbf{V}_o$ for $\mathbf{V}_1$.

$$h_{11}\mathbf{I}_1 + h_{12}\mathbf{V}_o + 1000\mathbf{I}_1 = \mathbf{V}_s$$

Substitute 500 for h_{11} and 10^{-4} for h_{12}.

$$500\mathbf{I}_1 + 10^{-4}\mathbf{V}_o + 1000\mathbf{I}_1 = \mathbf{V}_s$$

$$10^{-4}\mathbf{V}_o + 1500\mathbf{I}_1 = \mathbf{V}_s \quad \ldots\ldots (1)$$

Apply Kirchhoff's voltage law to the output loop.

$$\mathbf{I}_2 = h_{22}\mathbf{V}_o + h_{21}\mathbf{I}_1$$

Substitute 2×10^{-6} for h_{22} and 100 for h_{21}.

$$\mathbf{I}_2 = 2\times10^{-6}\mathbf{V}_o + 100\mathbf{I}_1$$

Also, $\mathbf{I}_2 = -\dfrac{\mathbf{V}_0}{2000}$

Substitute $-\dfrac{\mathbf{V}_0}{2000}$ for $\mathbf{I}_2$.

$$-\frac{\mathbf{V}_0}{2000} = 2\times10^{-6}\mathbf{V}_o + 100\mathbf{I}_1$$
$$100\mathbf{I}_1 = -\frac{\mathbf{V}_0}{2000} - 2\times10^{-6}\mathbf{V}_o$$
$$100\mathbf{I}_1 = -5.02\times10^{-4}\mathbf{V}_0$$
$$\mathbf{I}_1 = -5.02\times10^{-6}\mathbf{V}_0$$

Step 4 of 4

Substitute $-5.02\times10^{-6}\mathbf{V}_0$ for $\mathbf{I}_1$ in equation (1).

$$10^{-4}\mathbf{V}_o + (1500)\left(-5.02\times10^{-6}\mathbf{V}_0\right) = \mathbf{V}_s$$
$$10^{-4}\mathbf{V}_o - 7.53\times10^{-3}\mathbf{V}_0 = \mathbf{V}_s$$
$$-7.43\times10^{-3}\mathbf{V}_0 = \mathbf{V}_s$$
$$\frac{\mathbf{V}_0}{\mathbf{V}_s} = \frac{1}{-7.43\times10^{-3}}$$
$$\frac{\mathbf{V}_0}{\mathbf{V}_s} = -134.6$$

Thus, the ratio, $\dfrac{\mathbf{V}_0}{\mathbf{V}_s}$ is $\boxed{-134.6}$.